

Milestone 2 Real LFW Evaluation Report

Overview

This report evaluates the Milestone 2 face verification pipeline on the real LFW dataset already referenced by the repository. The baseline system uses deterministic grayscale-flattened features with cosine similarity. Thresholds are selected on the validation split by maximizing balanced accuracy, then locked for final reporting on the test split.

Tracked Runs And Threshold Rule

Five real tracked runs were completed to satisfy the milestone requirement, but the fairest baseline-versus-improved comparison is baseline_32 versus improved_32, because both use the same image size and scoring code while only the pair policy changes.

Run	Threshold	Val Balanced Acc.	Test Balanced Acc.	Test Acc.	Note
baseline_32	0.80	0.6210	0.5648	0.5648	Baseline reporting run
improved_32	0.77	0.6100	0.5852	0.5852	prefer_unique pair policy
improved_48	0.77	0.6112	0.5886	0.5886	Exploratory improved run

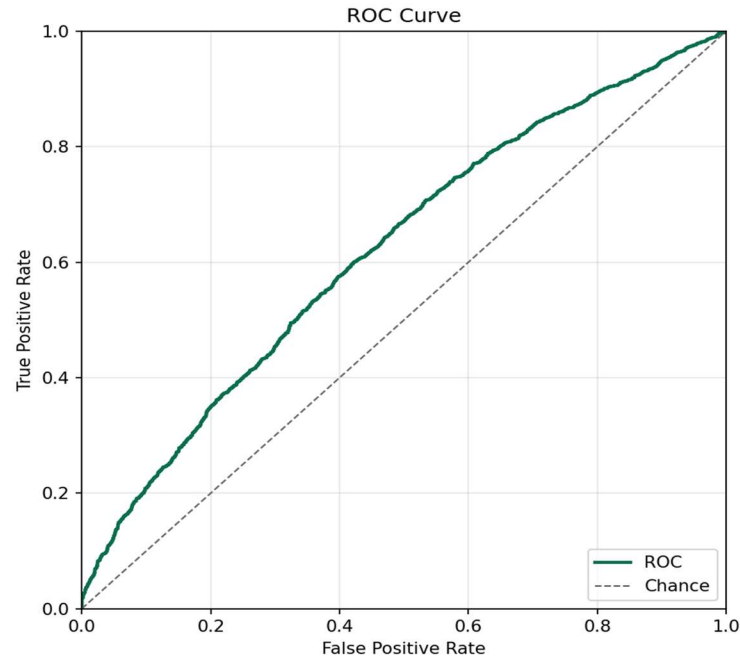
The threshold rule was fixed before final evaluation: choose the threshold on the validation split that maximizes balanced accuracy. For baseline_32, the selected threshold is 0.80. For improved_32, the selected threshold is 0.77.

Baseline Versus Improved Result

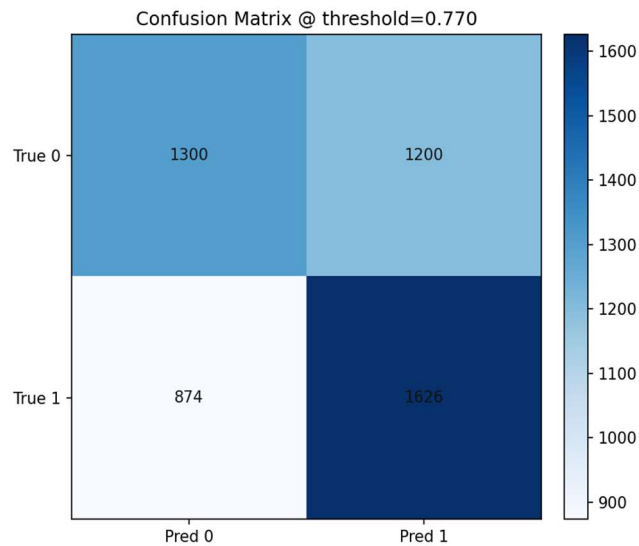
At the same 32x32 feature resolution, the improved pair policy raises test balanced accuracy from 0.5648 to 0.5852, an absolute gain of 0.0204. The main gain comes from recall: baseline recall is 0.4904, while improved recall is 0.6504. This comes with a tradeoff, because false positives increase from 902 to 1200.

ROC And Confusion Matrix

The improved 32x32 run is the clearest run to visualize because it is the real data-centric comparison target.



Improved 32x32 ROC plot



Improved 32x32 confusion matrix